MarketPulse: Visualizing Post-COVID Labor Market Dynamics

COM-480 Data Visualization — Milestone 2

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1. Project Goal

MarketPulse is an interactive web dashboard that explores post-COVID labor market recovery across OECD countries, with a focus on the extreme volatility in the U.S. information sector (tech, media, telecom). Central question: *How does international hiring resilience compare to U.S. information-sector volatility in job openings vs. layoffs?* The dashboard ingests three FRED time-series datasets: (i) Indeed Job Postings Index for 7 countries (AU, CA, DE, FR, GB, IE, US), (ii) U.S. Job Openings Rate (BLS JOLTS), and (iii) U.S. Layoffs & Discharges Rate. Target audience: job seekers, economists, HR professionals, and data journalists.

2. Visualization Design and Sketches

Figure 1 shows the running prototype with sidebar navigation, a top-bar global timeline filter, and a main content area. Figures 2–3 zoom into individual components.

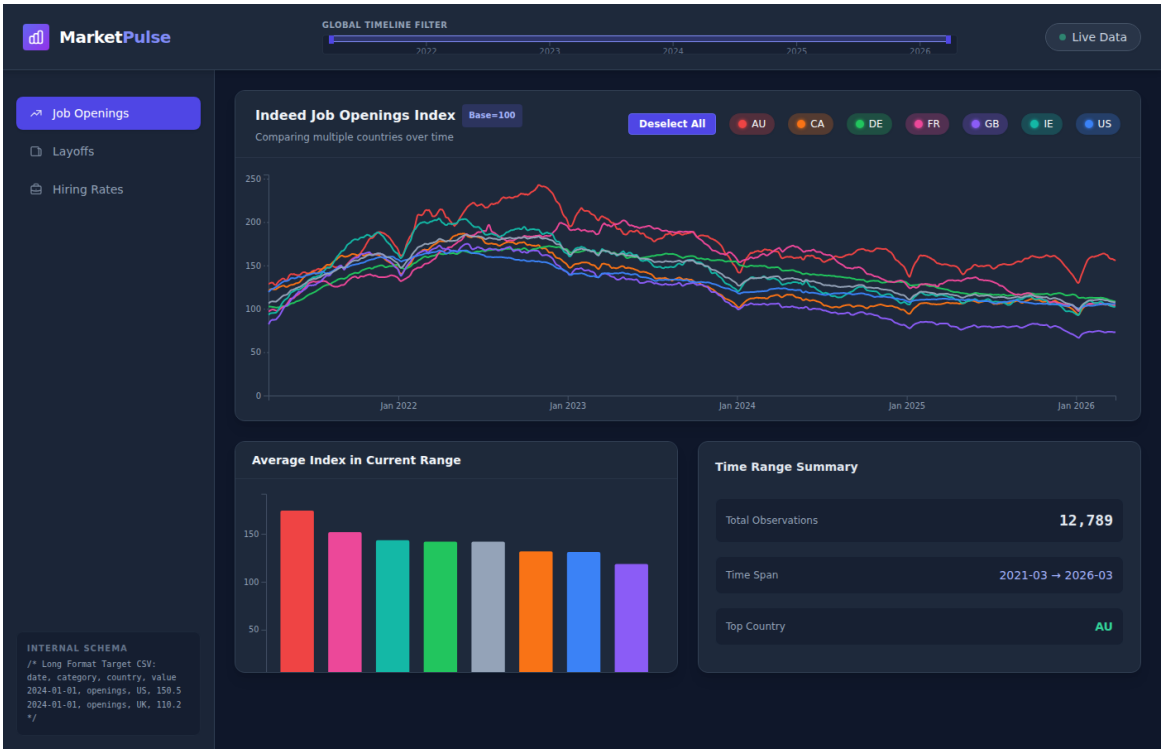

Figure 1: Full dashboard prototype — sidebar (left), global D3 brush timeline (top), multi-country line chart (center), bar chart and KPI summary panel (bottom).

Figure 2: Global `d3.brushX` timeline (zoomed) — users drag selection handles to filter all widgets across a shared time window.

Figure 3: Multi-series line chart — Indeed Job Postings Index across 7 countries with hover tooltips, color encoding, and adaptive "Average" trend.

The visualization is designed around the principle of *coordinated multiple views*: all widgets share a single reactive state, so brushing the timeline (Fig. 2) instantly updates the line chart (Fig. 3), the bar chart, and the summary KPIs. Country filter chips in the legend act as a second interaction layer, allowing users to isolate specific economies or compare subsets. An adaptive "Average" series is dynamically computed over the active selection, providing an at-a-glance benchmark. This linked-view architecture ensures that users can fluidly move between overview and detail without losing context, following Shneiderman's visual information-seeking mantra: *overview first, zoom and filter, then details-on-demand.*

3. Tools and Relevant Lectures

Component	Tools / Libraries	Relevant Lectures
Multi-series line chart	D3.js v7 (`d3.line`, `d3.scaleTime`, `d3.bisect`)	W4: D3.js, W11: Tabular Data
Bar chart	D3.js v7 (`d3.scaleBand`, `d3.rollup`)	W6: Marks & Channels
Brush / time filter	D3.js v7 (`d3.brushX`)	W5: Interaction, Adv. D3
Tooltips & hover	D3 pointer events, HTML tooltip	W6: Perception, Cognition
Dashboard layout	Tailwind CSS, CSS Grid / Flexbox	W7: Designing Viz (UI/UX)
Data pipeline & state	Vanilla JS (ES6+), `d3.csv`, pub-sub pattern	W2–3: JavaScript
Transitions / animation	`d3.transition`, `.duration()`	W4: D3.js, W5: Adv. D3
Scrollytelling (planned)	Scroll-linked D3 transitions	W13: Storytelling / Journalism

4. Implementation Breakdown

4.1 Core Visualization — Minimal Viable Product

Foundational features required for a complete, functional product:

- **Data pipeline & extensible registry:** Standardized CSV parser converting per-country files into unified long-format (`date, category, country, value`). New datasets added via config object. *Implemented.*
- **Reusable D3 chart factories:** Generic `createLineChart` and `createBarChart` accepting container selectors, data, and column mappings. *Implemented.*
- **Global brush time filter:** `d3.brushX` propagating time-range changes to all widgets via publish-subscribe state manager. *Implemented.*
- **Country toggle filters:** Chip-based legend with select/deselect and smooth enter/exit transitions. *Implemented.*
- **Hover tooltips with bisect snapping:** Nearest-point via `d3.bisect`, ranked per-country values with vertical crosshair. *Implemented.*
- **U.S. sector panels (Layoffs & Openings):** Dedicated views for JOLTS time series via the same chart factory. *Pending.*

4.2 Extra Ideas — Creative and Challenging Enhancements

These features elevate the dashboard into original territory but can be dropped without compromising the core:

- **Openings-vs-Layoffs phase-space scatter:** Animated scatter where each dot is a monthly (openings, layoffs) snapshot. Temporal trails reveal the boom→correction trajectory, inspired by Gapminder motion charts.
- **Annotated event timeline:** Contextual annotations ("COVID shock", "2022 peak", "2025 AI wave") overlaid on the line chart with labeled markers, turning raw data into guided narrative.
- **Scrollytelling guided tour:** Scroll-linked transitions with sticky chart and text panels, walking users through key insights with automated brush/filter changes.
- **Animated bar-chart race:** Month-by-month animated country ranking by job postings index, revealing which economies recovered fastest.
- **Volatility heatmap:** Calendar-style matrix encoding month-over-month volatility per country using a diverging color scale.
- **Responsive layout with micro-interactions:** Adaptive grid for mobile, with entrance animations and hover effects (bar glow, path morphing).

5. Functional Prototype

A working prototype is available on GitHub.[1] It includes: a dark-themed dashboard (Tailwind CSS), multi-country line chart with 8-series rendering (7 countries + adaptive average), summary bar chart, global D3 brush timeline, interactive country filter chips, KPI summary panel, and a pub-sub state architecture synchronizing all widgets. Data: real Indeed Job Postings Index (daily, 2021–2026) from CSV with smooth 500ms D3 transitions.

[1] https://github.com/com-480-data-visualization/Bakhano-Data-Viz